

RESEARCH ARTICLE

Hierarchical Reinforcement Learning for Navigation Among Movable and Immovable Obstacles

HAN JUN BAE¹ AND JUYOUN PARK^{1b2}, (Member, IEEE)

¹Samsung Electronics, Suwon 16677, Republic of Korea

²Center for Humanoid Research, Korea Institute of Science and Technology (KIST), Seoul 02792, Republic of Korea

Corresponding author: Juyoun Park (juyounpark@kist.re.kr)

This work was supported by Korea Institute of Science and Technology (KIST) Institutional Program under Project 2E33602.

ABSTRACT Path planning is a key technique for vehicle navigation, and significant research has focused on how to manage obstacles. Previous methods have addressed either removing movable obstacles or avoiding immovable ones. However, real-world environments contain both movable and immovable obstacles. To address this, we propose a path planning system for Navigation Among Movable and IMmovable Obstacles (NAMIMO) based on hierarchical reinforcement learning. In our system, lower-level agents generate paths for the vehicle to either avoid or remove obstacles, while the higher-level agent dynamically selects which lower-level agent to utilize based on the movability of the obstacle, incorporating visual and linguistic knowledge.

INDEX TERMS Navigation, local path planning, hierarchical reinforcement learning, visual and linguistic knowledge.

I. INTRODUCTION

Path planning is a crucial task for vehicles, enabling them to navigate to their destinations in dynamic environments. Global path planning generates the entire path from the starting point to the destination using a predefined map of the environment [1], [2], [3], [4]. Meanwhile, local path planning [5], [6], [7], [8], [9], [10], [11], [12] focuses on generating a local, feasible and safe path for the vehicle to follow in real time as it moves towards its goal. Local path planning algorithms utilize input from various sensors, such as cameras, light detection and ranging (LiDAR), and sonars, to build a local map of the environment. In local path planning, it is critical to deal with obstacles in dynamic and unpredictable environments, such as crowded public spaces or the addition of new obstacles.

Prior local path planning methods can be categorized into two groups: obstacle avoidance and obstacle removal. Obstacle avoidance methods [8], [9], [10], [11], [12] assume that the vehicle cannot relocate obstacles to clear its path

and thus generate a route to circumvent the obstacles. On the other hand, obstacle removal methods [5], [6], [7] concentrate on strategies for eliminating obstacles under the assumption that the vehicle can displace them. However, since real-life environments often comprise both movable and immovable objects, it is crucial not only to generate routes to avoid obstacles or devise obstacle removal strategies but also to determine beforehand whether the vehicle can relocate the obstacles.

Hence, in this paper, we propose a system for *Navigation Among Movable and IMmovable Obstacles (NAMIMO)* that selectively employs an obstacle removal or obstacle avoidance processes based on the movability of obstacles, using a Hierarchical Reinforcement Learning (HRL) approach. HRL is a framework designed to enhance the efficiency and scalability of Reinforcement Learning (RL) algorithms by organizing them into a hierarchical structure [13], [14], [15], [16], [17], [18], [19]. In HRL, tasks are composed of combinations of subtasks, with each subtask featuring its own RL agent (lower-level agent). Meanwhile, a higher-level agent determines the composition of these subtasks. Our proposed method involves training two lower-level agents,

The associate editor coordinating the review of this manuscript and approving it for publication was Zheng Chen^{1b}.

tasked with either obstacle avoidance or obstacle removal. And, the higher-level agent learns to interactively select a specific lower-level agent based on the movability of the obstacle currently encountered.

The most straightforward approach to assessing obstacle movability is by categorizing them. However, relying solely on object classification is insufficient for accurately determining obstacle movability. Factors such as whether an object has wheels can significantly influence its ability to move. Therefore, recognizing both the class of obstacles and their specific characteristics can enhance our understanding of obstacle movability. To extract not only object classes but also object properties from RGB image data, such as the presence of wheels, we utilize the Contrastive Language-Image Pre-Training (CLIP) features [20]. The CLIP model incorporates linguistic knowledge when analyzing the image. Consequently, features extracted from the CLIP model encompass not only visual information but also semantic object properties. This allows the agent to infer high-level attributes such as object category, material, or likely movability directly from visual input. CLIP's ability to capture such semantic object properties has been demonstrated through its zero-shot classification performance on benchmarks, where it outperformed prior models by an average of 2.6% using the ViT-L/14@336px backbone without task-specific training [20]. CLIP's strong generalization across diverse vision-language tasks has also been validated in subsequent studies, with further improvements achieved via prompt engineering and fine-tuning techniques [21], [22]. We integrate the CLIP encoder into the proposed navigation method and produce intermediate outputs fed to the higher-level agent to consider obstacle movability when determining the path planning strategy, whether it involves obstacle avoidance or removal. Through this process, even objects of the same type result in different paths being planned, primarily influenced by their properties, particularly their movability.

Moreover, our method, based on the HRL structure, can respond to obstacles with unpredictable movability, such as boxes with the same appearance but different weights. This capability stems from the proposed method, empowering the higher-level agent to dynamically re-decide which lower-level agent to select based on the performance of the current lower-level agent. For instance, if an obstacle is perceived as movable, the lower-level agent tasked with obstacle removal will be selected by our system to generate a path. Nonetheless, if the obstacle remains unremoved, this will be identified through the lower reward values yielded in RL from the lower-level agent. Subsequently, the higher-level agent will then switch to select the lower-level agent responsible for obstacle avoidance.

Our contributions can be summarized as follows:

- 1) We propose a system capable of path planning in environments with both movable and immovable objects, depending on the movability of the object.

- 2) We integrate the CLIP encoder with our proposed navigation model to incorporate linguistic information and consider the movability of objects.
- 3) We leverage the HRL structure to construct a unified framework that enables immediate activation of different path planning RL agents (obstacle avoidance or removal) and adapting to unpredictable obstacle movability.

II. RELATED WORK

A. LOCAL PATH PLANNING FOR OBSTACLE AVOIDANCE

Local path planning involves generating a sequence of motion commands that enable the vehicle to move from its current position to a desired goal while avoiding or removing obstacles in the environment. Obstacle avoidance methods have predominantly employed various types of RL algorithms, such as Proximal Policy Optimization (PPO) [8], Trust Region Policy Optimization (TRPO) [9], Asynchronous Advantage Actor-Critic (A3C) [10], Deep Deterministic Policy Gradient (DDPG) [11], and Twin Delayed Deep Deterministic policy gradient algorithm (TD3) [12]. These approaches typically penalize collisions with obstacles while rewarding successful navigation to the destination. Therefore, LiDAR information has been commonly used as input to obtain distance data from obstacles. On the other hand, several studies have explored the use of Convolutional Neural Network (CNN) features extracted from depth images as state representations for RL models. In [23], a depth image is generated from RGB data using a Generative Adversarial Network (GAN) model, and the image is then employed to train a PPO algorithm with a discrete action space to implement obstacle avoidance. In [24], the features derived from a sequence of depth images through convolutional layers are used as input for DDPG model to implement obstacle avoidance.

B. NAVIGATION AMONG MOVABLE OBSTACLES (NAMO)

Navigation Among Movable Obstacles (NAMO) is a challenging problem, as it involves planning trajectories for a vehicle to relocate the obstacles. Methods for obstacle removal typically entail global path planning based on a predefined map. While the vehicle traverses the planned path, its response to encountered obstacles varies depending on whether the obstacle can be moved or not. If an obstacle can be displaced, the vehicle clears the obstacle from the path and proceeds along its original trajectory. However, if the obstacle is immovable, the obstacle is added to the predefined map, prompting the vehicle to recalculate its path using global path planning techniques to find an alternative route. In [5], they estimate the shapes and sizes of obstacles based on point cloud information. If the size is below a certain threshold, the system determines that the vehicle has ability to remove the obstacle. They use a Contact Implicit Trajectory Optimization (CITO) method [25] to perform obstacle removal along the path. In [26], Deep Object Pose

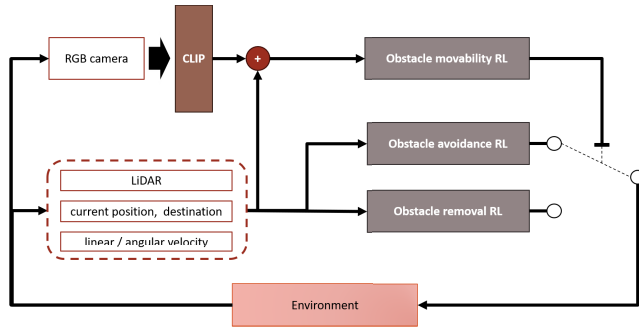


FIGURE 1. Proposed local path planning system. Obstacle movability RL decides which of obstacle avoidance RL and obstacle removal RL to plan the route based on information including CLIP features.

Estimation (DOPE) [13] based on RGBD information is employed to recognize obstacles, and the obstacle removal method used in S-NAMO [14] is utilized before proceeding along the original path.

C. HIERARCHICAL REINFORCEMENT LEARNING (HRL)

The primary objective of HRL is to tackle complex tasks by decomposing them into smaller, more manageable subtasks. HRL approach enables agents to learn and make decisions at multiple levels of abstraction, allowing them to address more challenging problems with less computational effort. One of the influential approaches is the H-DQN (hierarchical deep Q-network) [15], which extends the traditional Q-learning framework by introducing a hierarchy of policies. Other approaches based on the FeUdal Networks [16], [17], [18] present a method for learning hierarchies of policies where a higher-level manager communicates the subtask as a sub-goal, and a lower-level worker's objective is to reach the sub-goal. Additionally, other approaches based on the options frameworks [19], [27], [28] have provided valuable insights into decomposing complex tasks into a hierarchy of sub-tasks. These works with HRL enables agents to efficiently solve intricate problems by organizing actions into meaningful hierarchies and learning in complex environments. Our method draws inspiration from the HRL approach, enabling us to train both obstacle avoidance and obstacle removal tasks within a unified framework.

III. PROPOSED SYSTEM

We propose the NAMIMO system that enables navigation in environments featuring both movable and immovable objects by avoiding or relocating obstacles, depending on their movability, to reach a designated destination. Fig. 1 shows an overview of our proposed system. As depicted in the figure, we propose an HRL structure for the selective utilization of different RL algorithms, depending on the movability of obstacles. Our proposed HRL framework comprises three types of RL agents: obstacle avoidance RL for circumventing immovable objects and obstacle removal RL for displacing movable objects, acting as the lower-level agents, and movability RL for determining which RL agent

(obstacle avoidance RL/obstacle removal RL) to employ as the higher-level agent. Obstacle avoidance RL and obstacle removal RL leverage data from the vehicle, including LiDAR data, destination location, current position, linear velocity, and angular velocity, to control the vehicle's movements. Movability RL incorporates RGB information in addition to the data used for both obstacle avoidance RL and obstacle removal RL to consider the movability of obstacle. Specifically, the CLIP features extracted from images captured by RGB cameras convey the object's semantic properties indicating its movability. Obstacle movability RL uses these supplementary inputs to select one of the obstacle avoidance RL and obstacle removal RL path generation modules based on the object's movability.

A. OBSTACLE AVOIDANCE RL

For the vehicle to circumvent immovable obstacles, we utilize DDPG [29], which is designed to tackle problems in RL featuring continuous action spaces, for local path planning. DDPG integrates elements of deep neural networks and the actor-critic architecture. In order to train the DDPG-based local path planning for obstacle avoidance, we formulate the agent(vehicle)'s state space $\mathcal{S}^v$, action space $\mathcal{A}^v$, and the obstacle avoidance reward function R^a as follows.

State space $\mathcal{S}^v$: The agent's state $s^v \in \mathcal{S}^v$ is composed of the distance d^g and orientation θ^g between the agent and the goal, as well as the distance from the agent to the nearest obstacle d^o and the distance array data from LiDAR, $\mathbf{d}^l$. Each component is derived from sensor data collected from the IMU and LiDAR.

Action space $\mathcal{A}^v$: The agent takes an action $\mathbf{a}^v \in \mathcal{A}^v$ at each time step, transitioning from the current state to the next. The agent's action consists of the linear velocity $v \in [-1, 1]$ and the angular velocity $\omega \in [-1, 1]$. In the case of the obstacle avoidance RL agent, the vehicle is expected to learn to circumvent the immovable obstacle throughout training.

Reward function R^a : In obstacle avoidance RL, the following reward function is defined for the agent to learn the path of maneuvering to avoid obstacles.

$$R^a = \begin{cases} r_{success} & \text{if arrival} \\ r_{fail} & \text{if collision} \\ v' - |\omega'| - d^g & \text{otherwise} \end{cases} \quad (1)$$

where v' and ω' represent the agent's linear and angular velocities at the previous time step, respectively. d^g is the distance between the agent and the goal which encourages the agent to move closer to the goal. If the agent crashes into a wall or obstacles, the episode terminates immediately with a reward of $r_{fail} = -150$. If the agent reaches the destination, the episode ends immediately with a reward of $r_{success} = 150$.

B. OBSTACLE REMOVAL RL

Similar to the obstacle avoidance RL, we utilize the DDPG for the obstacle removal RL. We also employ the same state space $\mathcal{S}^v$ and action space $\mathcal{A}^v$ for the vehicle as those used

Model name	Screenshot	{s} object with the wheeled robots		a photo of a {c}	
Bevo Ottoman		movable	27.98 %	chair	91.99 %
		pushable	26.29 %	couch	3.57 %
		nonpushable	23.94 %	Table	3.00 %
Birch LowBackSeat		nonpushable	34.59 %	chair	89.55 %
		movable	25.32 %	couch	9.44 %
		immovable	22.69 %	bed	0.32 %

FIGURE 2. Example results of applying CLIP to chair images with the sentences '{s} object with the wheeled robots' where $s = \{\text{movable, pushable, nonpushable, immovable}\}$ and 'a photo of {c}' where c is a name of object.

in the obstacle avoidance RL. In obstacle removal RL, unlike in obstacle avoidance RL, the agent is trained to intentionally make contact with obstacles, i.e., collide with them, to move them while continuing to follow the original path. To achieve this, the obstacle removal reward function R^r is formulated differently from R^a .

Reward function R^r : For the obstacle removal task, collisions are intentional. Therefore, we remove the penalty for collisions from the reward function used in obstacle avoidance RL. The reward function is set as follows:

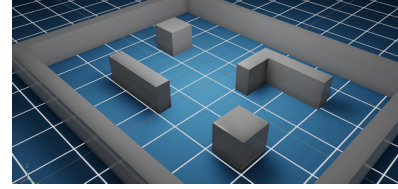
$$R^r = \begin{cases} r_{\text{success}} & \text{if arrival} \\ v' - |\omega'| - d^g - d^o & \text{otherwise} \end{cases} \quad (2)$$

where d^o represents the distance between the agent and the obstacle, encouraging the agent to approach the obstacle.

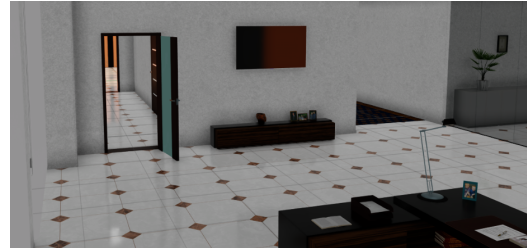
C. OBSTACLE MOVABILITY RL

In obstacle movability RL, the CLIP framework [20] is employed to assess the movability of objects based on their characteristics, facilitating selective choices for object avoidance and removal RL. The objective of CLIP is to align the embedding spaces of vision and language through a contrastive learning approach. By leveraging CLIP, we can calculate the similarities between the image and the text prompts in the embedding space. In our scenario, it can be utilized to ascertain the movability of the obstacle by measuring the similarity between the features of the obstacle image and the features of the sentence "the obstacle is movable." Fig. 2 displays the CLIP results with sample images of chairs provided by the NVIDIA Omniverse framework. As depicted in the figure, even within objects classified under the same category "chair," they can exhibit diverse attributes such as being movable, immovable, pushable, or non-pushable. This suggests that employing the CLIP framework can furnish information encompassing not only the object types but also their attributes when utilized as input for obstacle movability RL.

To offer comprehensive information, rather than explicitly determining movability in this manner, we develop an obstacle movability RL that utilizes CLIP (ViT-B/32 pretrained



(a) Simple environment



(b) Realistic environment

FIGURE 3. Screenshots of the simulation environments.

model [30]) features as input to implicitly ascertain the movability of the obstacle. Consequently, even when the object type remains the same but the movability varies due to different attributes, such as chairs with or without wheels, different path planning modules can be employed depending on the circumstances.

In selecting between obstacle avoidance RL and obstacle removal RL, which constitutes a discrete action, we employ DQN for obstacle movability RL. To account for the obstacle's movability, we incorporate CLIP features. Specifically, the state for obstacle movability RL s^m encompasses the CLIP features of the obstacle, along with the vehicle's state s^v utilized in both obstacle avoidance RL and obstacle removal RL. This enables the movability RL to consider the attributes of obstacles as well as the spatial relationship among the agent, the obstacles, and the goal.

For obstacle movability RL, we employ 0 or 1 as an action a^m to indicate the selective utilization of two types of RL: 0 for obstacle avoidance RL and 1 for obstacle removal RL. In the obstacle movability RL, reward values are determined based on the overall success of the navigation. Accordingly, the corresponding RL agent is trained by the rewards obtained from the selected module, either obstacle avoidance RL or obstacle removal RL. The reward function for obstacle movability R^m is defined as follows:

$$R^m = \begin{cases} R^a & \text{if select obstacle avoidance RL} \\ R^r & \text{if select obstacle removal RL.} \end{cases} \quad (3)$$

IV. EXPERIMENTS

A. EXPERIMENTAL SETUP

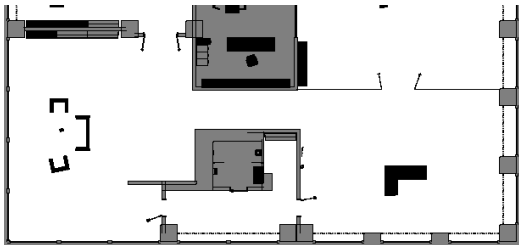
To validate the effectiveness of our proposed system, we conducted indoor navigation experiments with a mobile robot in simulation environments. We utilized the following two virtual environments within the NVIDIA Isaac Sim on top of the Omniverse platform: a simple environment (a)



FIGURE 4. JetBot model provided by NVIDIA Isaac SDK.



(a) Simple environment



(b) Realistic environment

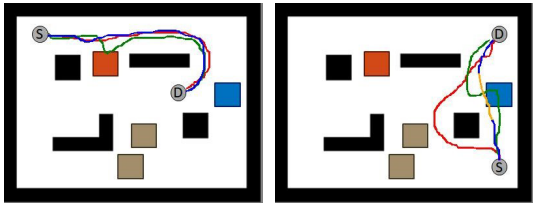
FIGURE 5. Global maps (white space: space where the mobile robot can traverse, black & gray space: space where the mobile robot cannot traverse).

Model name	Bevo Ottoman	Birch LowBackSeat	Cardbox B2
Type	Wheeled chair	Chair	Box
Screenshot			
Movability	movable	immovable	movable/immovable

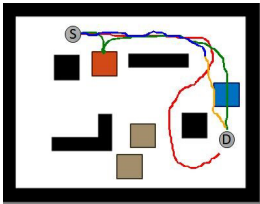
FIGURE 6. Properties of the obstacles used in the experiments.

and a realistic environment (b), as shown in Fig. 3. The simple environment was created with basic walls within a rectangular space, while the realistic environment was constructed by connecting two office rooms (provided by the Omniverse platform) with two corridors. Fig. 5 shows two global maps of the environments. For the mobile robot, we utilized the JetBot provided by the simulator (Fig. 4).

To replicate real-world environments with both movable and immovable obstacles, we employed various types of chairs and boxes. Fig. 6 displays screenshots and details regarding the characteristics of the chairs and boxes utilized as movable and immovable obstacles in the experiments. As depicted in the figure, wheeled chairs were set to be movable by the mobile robot, while chairs without wheels were designated as immovable. Moreover, despite having

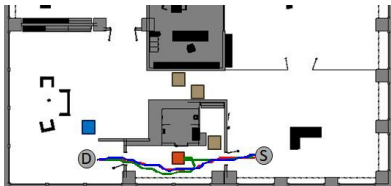


(a) Test Lap 1 (b) Test Lap 2

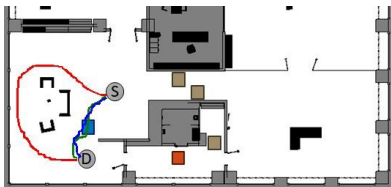


(c) Test Lap 3

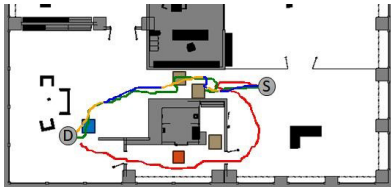
FIGURE 7. Travel paths of the robot for the simple environment experiment (red line: avoidance-only, green line: removal-only, blue line: proposed method (avoidance RL activated) yellow line: proposed method (removal RL activated), red box: immovable chair, blue box: movable chair, brown box: box, S: start point, D: destination).



(a) Test Lap 4



(b) Test Lap 5



(c) Test Lap 6

FIGURE 8. Travel paths of the robot for the realistic environment experiment (red line: avoidance-only, green line: removal-only, blue line: proposed method (avoidance RL activated) yellow line: proposed method (removal RL activated), red box: immovable chair, blue box: movable chair, brown box: box, S: start point, D: destination).

identical dimensions, the boxes were assigned different weights to enable distinct interactions with the robot: movable and immovable. To train our proposed model, we positioned obstacles such as chairs and boxes arbitrarily within each environment.

For comparative evaluation, we implemented two additional methods: 1) an obstacle avoidance system based on

TD3 model using LiDAR data [12], referred to as the ‘avoidance-only method’ in this paper, and 2) an obstacle removal system based on the A^* global path planning method assuming all obstacles are movable [26], referred to as the ‘removal-only method’ in this paper. These two methods can be considered as components of the proposed model. Depending on visual recognition, if the obstacle ahead is movable, the removal-only method is selected and activated; otherwise, the avoidance-only method is used. Therefore, this comparative experiment also serves as an ablation study. We selected TD3 and A as comparison baselines because they represent well-established approaches for obstacle avoidance and global planning in environments with exclusively movable or immovable objects, respectively. Since our setting involves both types of obstacles simultaneously, comparisons with semantic-based or hybrid methods developed for more specialized scenarios would be less directly applicable. Each model was trained with randomly assigned starting and destination points for each episode.

We employed the DDPG algorithm for policy learning, where both the Actor and Critic networks consisted of two fully connected hidden layers with 400 and 300 neurons, respectively, using ReLU activations. The learning rates were set to 0.0001 for the actor and 0.001 for the critic networks. A soft target update rate of 0.001 was used, and exploration during training was facilitated using an Ornstein-Uhlenbeck noise process with parameters $\theta = 0.15$, $\mu = 0.0$, and $\sigma = 0.2$. The input to the networks was constructed by concatenating heterogeneous observation sources, including: (i) downsampled CLIP image features representing high-level semantic visual cues, (ii) LiDAR-derived depth data indicating obstacle proximity, and (iii) scalar values for relative distance and orientation to the goal. This simple vector-level fusion was adopted to preserve computational efficiency, without employing additional cross-modal attention or fusion modules. The supplementary video contains the demonstrations.

B. EXPERIMENTAL RESULTS

Fig. 7 and 8 illustrates the starting points (S), destinations (D) of each test lap, and the travel paths (colored lines) for each algorithm across various environments. The red, green, blue, and yellow lines correspond to the paths traversed using the avoidance-only method, the removal-only method, the obstacle avoidance RL of the proposed method, and the obstacle removal RL of the proposed method, respectively. In contrast to conventional RL evaluations that emphasize average reward progression or episodic success rates, our evaluation focuses on trajectory efficiency—quantifying how directly and smoothly the agent navigates to the goal. This better reflects the practical demands of real-world navigation systems. The proposed method selectively employs obstacle avoidance (blue) or removal (yellow) depending on the scenario. To effectively demonstrate the possibility of efficient navigation through the generation of selective paths within a single experiment, distinct colors were employed.

TABLE 1. Travel distance (m) of the robot for each test lap.

Distance (m)	Avoidance-only method	Removal-only method	Proposed method
Test Lap 1	34.294	37.205	33.737
Test Lap 2	30.456	25.406	20.098
Test Lap 3	49.513	38.446	33.639
Test Lap 4	44.244	50.444	45.396
Test Lap 5	62.381	22.482	21.951
Test Lap 6	88.515	71.698	59.657

As the mobile robot navigates with the proposed method, the movability RL switches and activates either the avoidance RL or the removal RL. Consequently, in the path traversed with the proposed method, alternating blue (indicating avoidance RL activated) and yellow lines (indicating removal RL activated) are observed, depending on which path planning module is activated. Furthermore, some paths intersect with obstacles, indicating instances where the robot pushes obstacles to relocate and follow the original path.

Table 1 lists the travel distances of the robot using each algorithm for every test lap. As shown in the table, the proposed method achieved comparable distances to the avoidance-only method in Test Laps 1 and 4, while outperforming the removal-only method. Fig 7 (a) and 8 (a) depict the paths for each algorithm in Laps 1 and 4. As illustrated, the removal-only method planned its path assuming that obstacles were movable. Consequently, with this method, the robot attempted to relocate obstacles, and if no movement was detected for a few seconds, it incorporated the obstacle into the predefined map and recalculated the path using the global path planning method. Hence, the removal-only method resulted in a longer route. On the other hand, the proposed method performed the avoidance RL for obstacles such as walls and continued to circumvent when encountering immovable obstacles, allowing it to detour them immediately, resulting in a shorter distance compared to the removal-only method.

For Test Lap 2 and 5 in Table 1, the proposed method achieved comparable distance compared to the removal-only method, outperforming the avoidance-only method. Fig. 7 (b) and 8 (b) show the paths of each algorithm for Test Lap 2 and 5. As shown in the figures, the avoidance-only algorithm generated longer paths by avoiding the movable obstacle, even when it was possible to take a shorter route by removing the obstacle. This is because the avoidance-only method assumes that all obstacles are immovable. On the other hand, the proposed method conducted with the avoidance RL (blue lines in the figures), when the robot encountered obstacles like walls. And, when the robot detected a movable obstacle, our system switched to select the removal RL to relocate the obstacle (yellow lines in the figures) and remove it from the path. Afterward, the system resumed the avoidance RL to reach the destination (blue lines in the figures). This hybrid approach allows the proposed method to efficiently navigate around obstacles and reach the goal using a shorter path compared to the avoidance-only method.

For Test Lap 3 in Table 1 (travel paths in Fig. 7 (c)), the proposed method outperformed both the removal-only method and the avoidance-only method. In this scenario where both movable and immovable obstacles were present along the path to the destination, the advantages of the proposed method explained in the previous two paragraphs were observed, resulting in higher performance compared to both the removal-only method and the avoidance-only method.

For Test Lap 6 in Table 1 (travel paths in Fig. 8 (c)), the proposed method achieved comparable distance compared to the removal-only method, outperforming the avoidance-only method. In this scenario, the obstacles with unpredictable movability are present along the path to the destination. In this case, when the robot detected the obstacles, the proposed method tried to push the obstacle with removal RL (yellow line in the figure). But the removal RL did not work properly, the movability RL changed to conduct avoidance RL (blue line in the figure). As a result, as shown in the figure, our proposed system was able to selectively conduct avoidance RL or removal RL based on the situation through the movability RL, allowing it to efficiently handle scenario that even includes obstacles with unpredictable movability.

V. CONCLUSION AND FUTURE WORK

In this paper, we proposed a local path planning system that can handle both movable and immovable obstacles using an HRL approach. The proposed method selectively employs the DDPG-based obstacle avoidance or removal RL with the DQN-based obstacle movability RL. Through evaluations conducted in environments featuring both movable and immovable obstacles, our system demonstrated efficient navigation capabilities compared to the obstacle removal-only and obstacle avoidance-only methods. We plan to validate the proposed system through real-world deployments along with extensive experiment analyses. Additionally, we aim to assess the system's performance across different vehicles and various types of obstacles in diverse environments.

ACKNOWLEDGMENT

This work was conducted while Han Jun Bae was affiliated with Korea Institute of Science and Technology (KIST), Seoul 02792, Republic of Korea.

REFERENCES

- [1] E. W. Dijkstra, "A note on two problems in connexion with graphs," in *Edsger Wybe Dijkstra: His Life, Work, Legacy*. USA: ACM, 2022, pp. 287–290.
- [2] F. Duchoň, A. Babinec, M. Kajan, P. Beňo, M. Florek, T. Fico, and L. Jurišica, "Path planning with modified a star algorithm for a mobile robot," *Proc. Eng.*, vol. 96, pp. 59–69, Mar. 2014.
- [3] S. Karaman and E. Frazzoli, "Sampling-based algorithms for optimal motion planning," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 846–894, Jun. 2011.
- [4] J. D. Gammell, S. S. Srinivasa, and T. D. Barfoot, "Informed RRT: Optimal sampling-based path planning focused via direct sampling of an admissible ellipsoidal heuristic," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Sep. 2014, pp. 2997–3004.
- [5] M. Wang, R. Luo, A. Ö. Önel, and T. Padiş, "Affordance-based mobile robot navigation among movable obstacles," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2020, pp. 2734–2740.
- [6] D. I. Kim and G. S. Sukhatme, "Semantic labeling of 3D point clouds with object affordance for robot manipulation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2014, pp. 5578–5584.
- [7] M. Levihn, J. Scholz, and M. Stilman, "Planning with movable obstacles in continuous environments with uncertain dynamics," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2013, pp. 3832–3838.
- [8] Z. Wang, Z. Hu, Y. Yang, and Y. Yin, "Research on PPO algorithm in solving AUV path planning problems," in *Proc. 2nd Int. Seminar Artif. Intell., Netw. Inf. Technol. (AINIT)*, Oct. 2021, pp. 73–79.
- [9] X. Lu, Y. Cao, Z. Zhao, and Y. Yan, "Deep reinforcement learning based collision avoidance algorithm for differential drive robot," in *Proc. 11th Int. Conf. Intell. Robot. Appl.*, Newcastle, NSW, Australia. Cham, Switzerland: Springer, 2018, pp. 186–198.
- [10] P. Mirowski, R. Pascanu, F. Viola, H. Soyer, A. J. Ballard, A. Banino, M. Denil, R. Goroshin, L. Sifre, K. Kavukcuoglu, D. Kumaran, and R. Hadsell, "Learning to navigate in complex environments," 2016, *arXiv:1611.03673*.
- [11] Y. Dong and X. Zou, "Mobile robot path planning based on improved DDPG reinforcement learning algorithm," in *Proc. IEEE 11th Int. Conf. Softw. Eng. Service Sci. (ICSESS)*, Oct. 2020, pp. 52–56.
- [12] R. Cimurs, I. H. Suh, and J. H. Lee, "Goal-driven autonomous exploration through deep reinforcement learning," *IEEE Robot. Autom. Lett.*, vol. 7, no. 2, pp. 730–737, Apr. 2022.
- [13] J. Tremblay, T. To, B. Sundaralingam, Y. Xiang, D. Fox, and S. Birchfield, "Deep object pose estimation for semantic robotic grasping of household objects," 2018, *arXiv:1809.10790*.
- [14] B. Renault, J. Saraydaryan, and O. Simonin, "Towards S-namo: Socially-aware navigation among movable obstacles," in *Proc. 23rd Robot World Cup (RoboCup)*. Cham, Switzerland: Springer, 2019, pp. 241–254.
- [15] T. D. Kulkarni, K. Narasimhan, A. Saeedi, and J. B. Tenenbaum, "Hierarchical deep reinforcement learning: Integrating temporal abstraction and intrinsic motivation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 3682–3690.
- [16] P. Dayan and G. E. Hinton, "Feudal reinforcement learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 5, 1992, pp. 271–278.
- [17] O. Nachum, S. Gu, H. Lee, and S. Levine, "Data-efficient hierarchical reinforcement learning," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 1–11.
- [18] A. S. Vechnyevets, S. Osindero, T. Schaul, N. Heess, M. Jaderberg, D. Silver, and K. Kavukcuoglu, "Feudal networks for hierarchical reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2017, pp. 3540–3549.
- [19] R. S. Sutton, D. Precup, and S. Singh, "Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning," *Artif. Intell.*, vol. 112, nos. 1–2, pp. 181–211, Aug. 1999.
- [20] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.
- [21] K. Zhou, J. Yang, C. C. Loy, and Z. Liu, "Learning to prompt for vision-language models," *Int. J. Comput. Vis.*, vol. 130, no. 9, pp. 2337–2348, 2022.
- [22] C. Ju, T. Han, K. Zheng, Y. Zhang, and W. Xie, "Prompting visual-language models for efficient video understanding," in *Proc. Eur. Conf. Comput. Vis.*, 2022, pp. 105–124.
- [23] P. Wenzel, T. Schon, L. Leal-Taixe, and D. Cremers, "Vision-based mobile robotics obstacle avoidance with deep reinforcement learning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2021, pp. 14360–14366.
- [24] R. Cimurs, J. H. Lee, and I. H. Suh, "Goal-oriented obstacle avoidance with deep reinforcement learning in continuous action space," *Electronics*, vol. 9, no. 3, p. 411, Feb. 2020.
- [25] J.-P. Sleiman, J. Carius, R. Grandia, M. Wermelinger, and M. Hutter, "Contact-implicit trajectory optimization for dynamic object manipulation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Nov. 2019, pp. 6814–6821.
- [26] K. Ellis, H. Zhang, D. Stoyanov, and D. Kanoulas, "Navigation among movable obstacles with object localization using photorealistic simulation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2022, pp. 1711–1716.
- [27] E. Brunskill and L. Li, "PAC-inspired option discovery in lifelong reinforcement learning," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 316–324.

- [28] P. Bacon, J. Harb, and D. Precup, "The option-critic architecture," in *Proc. AAAI Conf. Artif. Intell.*, vol. 31, 2017, pp. 1726–1734.
- [29] T. P. Lillicrap, J. J. Hunt, A. Pritzel, N. Heess, T. Erez, Y. Tassa, D. Silver, and D. Wierstra, "Continuous control with deep reinforcement learning," 2015, *arXiv:1509.02971*.
- [30] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.



HAN JUN BAE received the Ph.D. degree in electrical and electronic engineering from Korea University, Seoul, Republic of Korea, in 2022. From 2022 to 2024, he was a Postdoctoral Researcher with Korea Institute of Science and Technology (KIST), Seoul. He is currently with the SmartThings Server Research and Development Group, Device Platform Center, Samsung Electronics, Suwon, Republic of Korea. His research interests include deep learning applications in embedded systems, such as indoor localization for mobile users and intelligent control of mobile robots. He is also interested in deep learning-based the Internet of Things (IoT) applications and the optimization of shared CPU caches.



JUYOUN PARK (Member, IEEE) received the B.S. and Ph.D. degrees in electrical engineering from Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Republic of Korea, in 2015 and 2019, respectively.

In 2019, she was a Postdoctoral Researcher with the Information and Electronics Research Institute, KAIST. In 2020, she was a Postdoctoral Scientist with the Department of Biomedical Engineering, George Washington University, Washington, DC, USA. Since 2021, she has been a Senior Researcher with the Center for Humanoid Research, Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea. Her research interests include but are not limited to the areas of artificial intelligence (AI) for autonomous agents, including robots. In particular, she is interested in developing novel machine learning or deep learning methods to allow robots to autonomously recognize and understand their environments. She is interested in developing a framework for practical application of the developed intelligence to various robots, such as social robots and surgical assistive robots, and in industrial fields.

• • •